

(1001CJA102120056)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

FULL SYLLABUS

10-03-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)**ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	B	B	D	D	D	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	C	A	C	C	A	C	A	D
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	171.35 to 171.45	41.20 to 41.30	14.40	4.79 to 4.82	100.00	25.14	7.00	1.33	0.80	4.40 to 4.41

PART-2 : CHEMISTRY

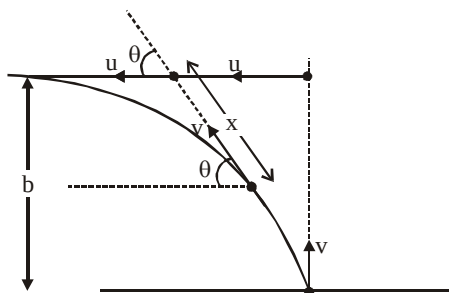
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	A	D	D	A	A	B	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	B	D	A	C	A	B	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5.00	7.00	4.00	4.08	0.20	16.60 to 16.70	0.11 to 0.12	5.00	5.00	4.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	C	B	B	A	B	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	C	A	B	B	C	C	A	C	B
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	13.00	6.00	16.00	1.00	4.00	1.00	4.00	3.00	1.00	5.00

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)**PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (A)**

Sol. Assume that water in river is at rest so Dock will move towards left with speed u & boat is continuously aiming at dock.



$$(v - u \cos \theta) = -\frac{dx}{dt}$$

$$-\int_b^0 dx = \int_0^T (v dt - u \cos \theta dt)$$

initial separation is b .

$$b = vT - u \int_0^T \cos \theta dt \quad \dots(1)$$

$\therefore \theta$ is variable.

$$uT = \int_0^T v \cos \theta dt$$

horizontal distance covered is same.

$$uT = v \int_0^T \cos \theta dt$$

$$\frac{uT}{v} = \int_0^T \cos \theta dt \quad \text{put in equation (i)}$$

$$b = vT - u \cdot \frac{uT}{v}$$

$$b = T \frac{(v^2 - u^2)}{v}$$

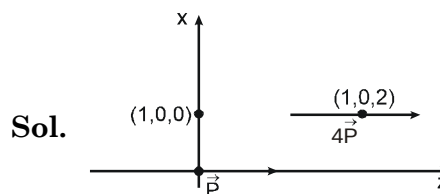
$$T = \frac{bv}{v^2 - u^2}$$

2. Ans. (A)

Sol. First kinetic friction acts down the incline then static friction acts up the incline.

3. Ans. (C)

Sol. For object to cross region of B and D it needs total energy more than +35 J

4. Ans. (B)**Sol.**

$$E_z = -\frac{kP}{(1)^3} + \frac{2k(4P)}{(2)^3} = 0$$

5. Ans. (B)

$$\text{Sol. } C = \frac{\epsilon_0}{d} [2A + (k-1)x]$$

$$q = CV$$

$$i = \frac{dq}{dt} = V \frac{dc}{dt}$$

6. Ans. (D)

Sol. Let end corrections at A and B α and β

$$\frac{15}{42+\alpha} = \frac{20}{58+\beta} \Rightarrow \frac{58+\beta}{42+\alpha} = \frac{4}{3}$$

$$\Rightarrow 3\beta + 174 = 168 + 4\alpha$$

$$4\alpha - 3\beta = 6 \quad \dots(i)$$

$$\frac{20}{57+\alpha} = \frac{15}{43+\beta} \Rightarrow \frac{57+\alpha}{43+\beta} = \frac{4}{3}$$

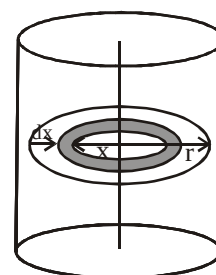
$$= \frac{4}{3} \Rightarrow 171 + 3\alpha = 172 + 4\beta$$

$$3\alpha - 4\beta = 1 \quad \dots(ii)$$

from (i) and (ii)

$$\alpha = 3 \text{ cm}, \beta = 2 \text{ cm}$$

$\therefore$ (4)

7. Ans. (D)**Sol.**

$$\oint B d\ell = \mu_0 I_{en}$$

$$\oint B_0 r^3 d\ell = \mu_0 I_{en}$$

or $B_0 r^3 \int_0^{2\pi} d\ell = \mu_0 I_{en}$ [Note : $B_0 r^3$ is taken

out of integration because it is not varying along $d\ell$]

or $B_0 r^3 (2\pi r) = \mu_0 I_{en}$ or $I_{en} = \frac{2\pi B_0 r^4}{\mu_0}$

8. Ans. (D)

Sol. By work-energy theorem;

$$qE_0 \cdot x = \frac{1}{2} m (13)^2 \Rightarrow x = \frac{169m}{2qE_0}$$

9. Ans. (D)

Sol. $|V| = \left| -\frac{d\phi}{dt} \right| = \oint \vec{E} \cdot d\vec{l} \Rightarrow \pi b^2 \frac{B}{\Delta t} = E \times 2\pi a$

$\therefore$ Force on circular wheel

$$= E \times q = E \times 2\pi a \times \lambda = \frac{\pi b^2 B}{\Delta t} \times \lambda$$

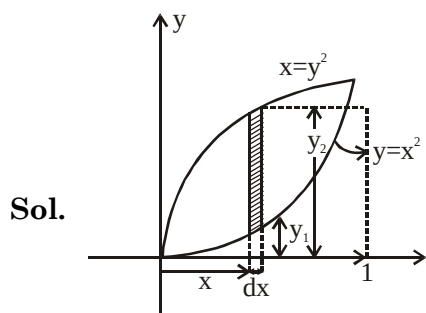
Torque about centre

$$= a \times F = \frac{\pi b^2 B}{\Delta t} \times \lambda a = I_C \alpha$$

$$\therefore I_C \times \frac{\omega}{\Delta t} = \frac{\pi b^2 B \lambda a}{\Delta t}$$

$$\therefore \omega = \frac{\pi b^2 B \lambda a}{I_C}$$

10. Ans. (A)

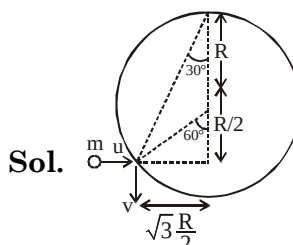


$$x_{cm} = \frac{\int x dm}{\int dm} = \frac{\int_0^1 x \lambda (y_2 - y_1) dx}{\int_0^1 \lambda (y_2 - y_1) dx}$$

$$= \frac{\int_0^1 (\sqrt{x} - x^2) x dx}{\int_0^1 (\sqrt{x} - x^2) dx} = \frac{9}{20}$$

$$y_{cm} = \frac{\int y dm}{\int dm} = \frac{\int_0^1 y \lambda (x_2 - x_1) dy}{\int_0^1 \lambda (x_2 - x_1) dy} = \frac{9}{20}$$

11. Ans. (B)



$$mu \times \frac{3R}{2} = mv \times \frac{\sqrt{3}R}{2} + \frac{3}{2} m R^2 \omega$$

$$3u = \sqrt{3}v + 3R\omega \quad \dots(i)$$

$$u \sin 30^\circ = v \sin 60^\circ$$

$$u = \sqrt{3}v \Rightarrow v = \frac{u}{\sqrt{3}} \quad \dots(ii)$$

$$3u = \sqrt{3} \frac{u}{\sqrt{3}} + 3R\omega$$

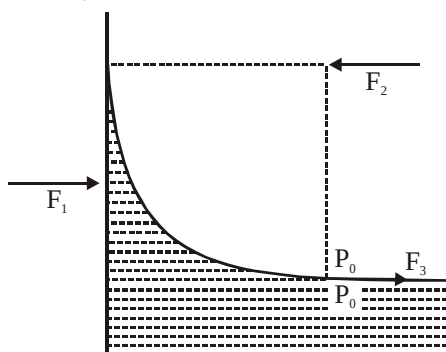
$$\omega = \frac{2u}{3R}$$

12. Ans. (B)

Sol. $F_1 + F_3 = F_2$

$$F_1 = \left(P_0 - \frac{\rho gh}{2} \right) h \ell$$

$$F_2 = P_0 h \ell$$

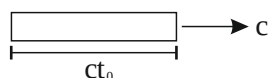


$$\left(P_0 - \frac{\rho gh}{2} \right) h \ell + T \ell = P_0 h \ell$$

$$h^2 = \left(\frac{2T}{\rho g} \right)$$

13. Ans. (C)

Sol. Let t_0 is the duration of beep. The length of air column disturbed in time t_0 is ct_0 .



$\frac{ct_0}{c+v_0}$ = duration for which observer listens, where v_0 is velocity of observer.

So, $\frac{ct_0}{c+v_0} \times v_0$ = distance over which observer listens.

$$\frac{c t_0 v_0}{c + v_0} = 78 \quad \text{----- (i)}$$

$$f = f_0 \left[\frac{c+v_0}{c} \right]$$

$$v_0 = 16.37 \text{ m/s}$$

Put in eq (i) $t_0 = 5 \text{ sec}$

14. Ans. (A)

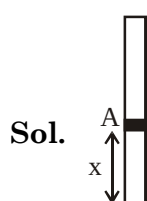
Sol. Angular width of central maxima

$$= \frac{2\lambda}{d} = \frac{2 \times 500 \times 10^{-9}}{10 \times 10^{-6}} = 0.1 \text{ rad}$$

$$= 5.75 \text{ degree}$$

$\therefore$ 2.86 degree on both side of central maxima which is at $\theta = 30^\circ$

15. Ans. (C)

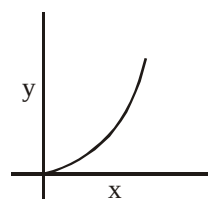


Sol.

Elastic energy density at

$$A = \frac{1}{2} \frac{(\text{stress})^2}{Y} = \text{constant}$$

$$\frac{1}{2} \frac{\left(\frac{M}{L} \frac{x}{A} g \right)^2}{Y} = \text{constant}$$



$$Y \propto x^{2+}$$

16. Ans. (C)

$$\text{Sol. } \frac{dQ}{dt} = \frac{\Delta T}{\frac{L}{KA'}} = \Delta T \frac{KA'}{L}$$

$$dQ = dU + dw$$

$$dU = 0$$

$$\frac{dQ}{dt} = \frac{dw}{dt} = PA \frac{dx}{dt} = PA v$$

$$\frac{\Delta TKA'}{L} = PA v$$

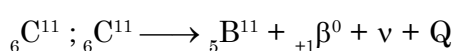
$$\Delta T = 27 \text{ K}, T = 300 \text{ K}$$

$$\frac{27K \times 1}{9L} = 0.5 \frac{RT}{V} Av$$

$$\text{Solving } v = \frac{K}{100R}$$

17. Ans. (A)

Sol. Equation of β^+ -decay of



Q-value of reaction = Δmc^2

$$= [m({}_6^{11}\text{C}) - 6m_e - m({}_5^{11}\text{B}) + 5m_e - m_e] c^2$$

$$= [m({}_6^{11}\text{C}) - m({}_5^{11}\text{B}) - 2m_e] c^2$$

$$= [11.011434 - 11.009305 - 2 \times 0.000548] \text{uc}^2$$

$$= [0.001033] \text{uc}^2 = 0.001033 \times 931.5 \text{ MeV}$$

$$= 0.962 \text{ MeV}$$

18. Ans. (C)

Sol. Since diode in forward bias so it behaves as short circuit.

$\therefore$ equivalent resistance of circuit

$$= \left(10 + \frac{10 \times 10}{10 + 10} \right) \text{ k}\Omega = 15 \text{ k}\Omega$$

$$\therefore \text{ current through cell} = \frac{30 \text{ V}}{15 \text{ k}\Omega} = 2 \text{ mA}$$

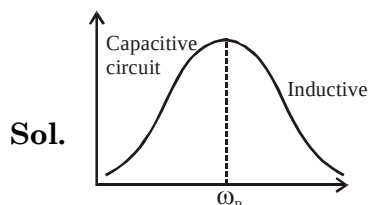
$$\therefore \text{ current between A and B} = 1 \text{ mA}$$

$$\therefore V_{AB} = 10 \text{ k}\Omega \times 1 \text{ mA} = 10 \text{ V}$$

19. Ans. (A)

Sol. $\frac{50g}{A} = \gamma \frac{\Delta L}{L}$

20. Ans. (D)



$$\omega_R = \frac{1}{\sqrt{LC}}$$

$\omega_2 < \omega_R$ from diagram

Circuit will behave as capacitance circuit

∴ Current leads voltage

SECTION-II

1. Ans. 171.35 to 171.45

Sol. Efficiency is given by

$$\eta = 1 - \frac{T_2}{T_1}$$

Here $\eta = 30$ and $T_2 = 300$ K, $T_1 = ?$

$$0.30 = 1 - \frac{300}{T_1}$$

or $T_1 = 428.6$ K

For increased efficiency of 50%, the source temperature should be

$$0.5 = 1 - \frac{300}{T_1}$$

or $T_1 = 600$ K.

Hence the Source Temperature should be increased by $600 - 428.6 = 171.4$ K

2. Ans. 41.20 to 41.30

Sol. $\frac{60 - P}{60} = 0.6 \left(1 - \frac{313}{653} \right)$

$$\eta_{\text{engine}} = \frac{60 - P}{60}$$

$$\eta_{\text{carnot}} = 1 - \frac{313}{653}$$

3. Ans. 14.40

Sol. Energy emitted by bulb/s = $100 \times 0.05 = 5$ J

Energy incident of one K atom in 1 s

$$= \frac{5}{4\pi R^2} \times \pi r^2 = \frac{5 \times 100^2}{4 \times 75^2} \times (10^{-10})^2$$

Time required for each atom to absorb 2eV

$$\text{energy} = \frac{2 \times 1.6 \times 10^{-9}}{\text{Energy incident in 1sec}} = 14.4 \text{ s}$$

4. Ans. 4.79 to 4.82

Sol. Total energy falling on the plate per unit area

$$= \frac{1}{4\pi(2)^2}$$

$$\text{No. of Photons hitting} = \frac{\frac{1}{4\pi(2)^2}}{\frac{1240 \times 1.6 \times 10^{-19}}{480}}$$

$$\Rightarrow 4.81 \times 10^{16}$$

5. Ans. 100.00

Sol. Amount of heat required to take ice + water mixture at 100°C .

$$((250 + 50) \times 1 \times 100) + (200 \times 80 + 200 \times 1 \times 100) = 66,000 \text{ cal.}$$

But the amount of energy released by steam while coming to 100°C water is

$$200 \times 540 = 108,000.$$

Which is more than heat required

∴ final temperature is 100°C and some part of steam will remain as it is.

6. Ans. 25.14

Sol. Heat absorbed by water and container

$$(100 + 200) \times 1 \times (T - 20)$$

Heat absorbed by ice

$$+ \left[40 \times \frac{1}{2} \times (20) \right] + 40 \times 1 \times (T) + 40 \times 80$$

Heat released by steam

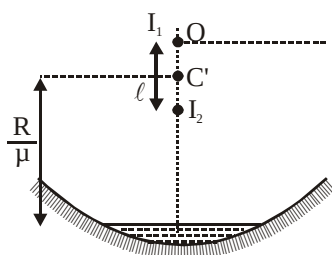
$$= 10 \times 540 + 10 \times 1 (100 - T)$$

Heat absorbed = Heat released.

Solving it $T = 25.14$

7. Ans. 7.00

Sol.



On pouring water the new center of curvature is at R/μ .

$$\frac{1}{v} + \frac{1}{-R} = \frac{2\mu}{-R}$$

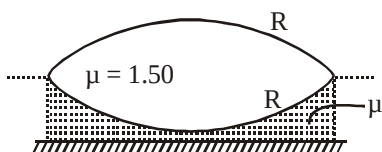
$$\therefore v = \frac{R}{1-2\mu} \text{ for } I_2; v = -5$$

$$\mu = 1.4$$

8. Ans. 1.33

• P

Sol.



Focal length of convex lens

$$\frac{1}{f} = (\mu - 1) \left[\frac{1}{R} - \frac{1}{(-R)} \right]$$

$$f = R$$

focal length of lens made of liquid

$$\frac{1}{f'} = (\mu - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right)$$

$$f' = \frac{-R}{\mu - 1}$$

case-2

$$\frac{1}{f'_{eq}} = \frac{1}{f_m} - \frac{1}{2f_L} \Rightarrow f'_{eq} = -2f_L = -2R [\because f_m \rightarrow \infty]$$

$$\Rightarrow f'_{eq} = -\frac{30}{2} = -2R \Rightarrow R = 7.5$$

case-1

$$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{1}{2f} - \frac{1}{2f'}$$

$$\Rightarrow \frac{1}{f_{eq}} = -\frac{1}{2R} + \frac{1}{2R(\mu - 1)}$$

$$\Rightarrow -\frac{2}{45} = \frac{1}{2R} [-1 + \mu - 1]$$

$$\Rightarrow -\frac{30}{45} = (\mu - 2) \Rightarrow \mu = 2 - \frac{2}{3}$$

$$\Rightarrow \mu = \frac{4}{3}$$

9. Ans. 0.80

Sol. Least count

= pitch / no. of division on circular scale

$$= 0.5 \text{ mm} / 50$$

$$= 0.01 \text{ mm}$$

Negative zero error = $-5 \times \text{LC} = -5 \times 0.01$

$$= -0.05 \text{ mm}$$

Measured value

= main scale reading + screw gauge reading
- zero error

$$= 0.5 \text{ mm} + 25 \times 0.01 \text{ mm} - (-0.05 \text{ mm})$$

$$= 0.75 \text{ mm} + 0.05 \text{ mm} = 0.80 \text{ mm}$$

10. Ans. 4.40 to 4.41

$$\text{Sol. Least count} = \frac{0.5 \text{ mm}}{100}$$

$$\text{Positive error} = 2 \times \frac{0.5 \text{ mm}}{100} = \frac{1}{100} \text{ mm}$$

$$\text{Scale reading} = 8 \times 0.5 \text{ mm}$$

$$+ 83 \times \frac{0.5 \text{ mm}}{100} - \frac{1}{100} \text{ mm}$$

$$\Rightarrow 4.405 \text{ mm}$$

SECTION-I

1. **Ans.(C)**

Acidity of the oxides of non-metals increases with the electronegativity and oxidation number of the element.
($\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_3$)
 $\text{B} < \text{C} < \text{O} < \text{N}$ (IE_1)

2. **Ans.(D)**

Salicylic acid $\rightarrow$ intramolecular hydrogen bonding

3. **Ans.(A)**

$[\text{Co}(\text{en})_2\text{Cl}_2]^+ \rightarrow$ Geometrical and optical isomerism

$[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+} \rightarrow$ No geometrical isomerism

$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+ \rightarrow$ Geometrical isomerism

$[\text{Cr}(\text{Ox})_3]^{3-} \rightarrow$ Optical isomerism

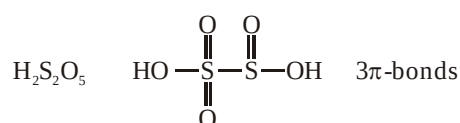
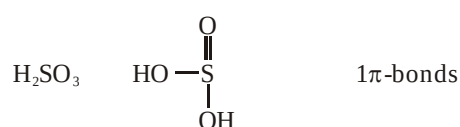
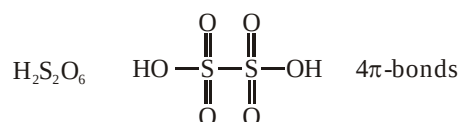
4. **Ans.(D)**

5. **Ans.(D)**

$\text{Be} < \text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$

(Reducing property in aqueous medium)

6. **Ans.(A)**



7. **Ans.(A)**

Refer. NCERT (d & f-block)

8. **Ans.(B)**

9. **Ans.(D)**

10. **Ans.(A)**

11. **Ans.(B)**

In case of conc. H_2SO_4 & conc. H_3PO_4 , E_1 mechanism will be followed ring expansion will occur but with POCl_3 / Pyridine E_2 mechanism.

12. **Ans.(C)**

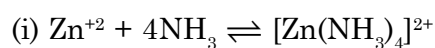


13. **Ans.(B)**

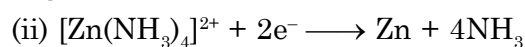
14. **Ans.(D)**

15. **Ans.(A)**

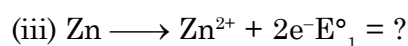
16. **Ans.(C)**



$$K_{\text{eq.}} = 10^9$$



$$E^\circ = -1.03 \text{ V}$$



Target equation

Equation (i) = -Equation (ii) - Equation (iii)

$$\Delta G_1^\circ = \Delta G_2^\circ - \Delta G_3^\circ$$

$$-RT \ln Q = +2 \times F \times (-1.03) - (-2FE^\circ_1)$$

$$-2.303 \times RT \log 10^9 = -2.06 F + 2FE^\circ_1$$

$$\left(-\frac{2.303RT}{F} \right) \times 9 = -2.06 + 2E^\circ_1$$

$$-0.059 \times 9 = -2.06 + 2E^\circ_1$$

$$E^\circ_1 = (1.03 - 4.5 \times 0.059) \text{ V}$$

$$0.77 \text{ V}$$

17. **Ans.(A)**

In experiments I and II initial concentrations of B and C remain unchanged and hence from the rate expression

$$r = k[A_0]^a[B_0]^b[C_0]^c = k_0[A_0]^a$$

$$\text{We get } \frac{(r_0)_1}{(r_0)_2} = \frac{2.08 \times 10^{-3}}{1.04 \times 10^{-3}} = \frac{[A_0]_1^a}{[A_0]_2^a} = \frac{0.02^a}{0.01^a}$$

$$\text{or } 2 = (a)^a \text{ and } a = 1$$

Similarly

$$\frac{(r_0)_3}{(r_0)_1} = \frac{4.16 \times 10^{-3}}{2.08 \times 10^{-3}} = \frac{[B_0]_3^b}{[B_0]_1^b} = \frac{0.04^b}{0.02^b}$$

$$2 = (2)^b \text{ or } b = 1$$

$$\text{and } \frac{(r_0)_4}{(r_0)_1} = \frac{8.32 \times 10^{-3}}{2.08 \times 10^{-3}} = \frac{[C_0]_4^c}{[C_0]_1^c} = \frac{0.04^c}{0.02^c}$$

$$\text{or } 4 = (2)^c \text{ and } c = 2$$

Therefore, the order of the reaction with respect to A, B and C are 1, 1 and 2 respectively and the overall order of the reaction = 1 + 1 + 2 = 4.

18. **Ans.(B)**

19. **Ans.(C)**

$$[Ag^+][I^-] = K_{sp}[AgI]$$

$$[Ag^+] = \frac{10^{-17}}{10^{-2}} = 10^{-15}$$

20. **Ans.(C)**

SECTION-II

1. **Ans.(5.00)**

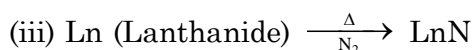
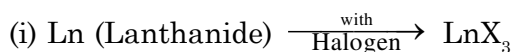
Na, K, Zn, Ca, Mg

2. **Ans.(7.00)**

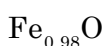
Classical smog → Chemically it is reducing mixture so called as reducing smog.

Photochemical smog → It has high concentration of oxidising agents so it is called as oxidising smog.

3. **Ans.(4.00)**



4. **Ans.(4.08)**



$$2\left(\frac{100-x}{100} \times 0.98\right) + 3\left(\frac{x}{100} \times 0.98\right) - 2 \times 1 = 0$$

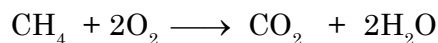
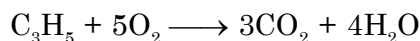
$$200 - 2x + 3x$$

$$200 + x = \frac{200}{0.98}$$

$$x = 4.08\%$$

5. **Ans.(0.20)**

$p \propto n$ at constant v & T



$$\text{Total CO}_2 \text{ pressure} = 3p + 320 - p = 448$$

$$2p = 128$$

$$p = 64$$

$$\text{mole fraction of C}_3\text{H}_8 = \frac{64}{320} = 0.2$$

6. **Ans. (16.60 to 16.70)**

$$\Delta S_{\text{sys}} = -nR \ln \frac{P_1}{P_2} = 10 \times 8.314 \ln \frac{1}{2}$$

$$= 10 \times 8.314 \times 0.7$$

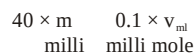
$$\Delta S_{\text{surrounding}} = -\frac{P\Delta V}{T} = \frac{-1\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right)}{T}$$

$$= -10 \times 8.314 \left(\frac{1}{1} - \frac{1}{2}\right)$$

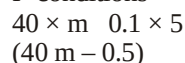
$$\Delta S_{\text{total}} = -10 \times 8.314 \times 0.7 + -10 \times 8.314 \left(\frac{1}{1} - \frac{1}{2}\right)$$

$$= 16.63$$

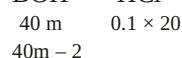
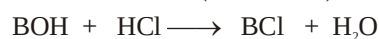
7. **Ans.(0.11 to 0.12)**



Ist conditions



$$\text{pOH} = \text{pKb} + \log \frac{[0.5]}{(40m - 0.5)} = 14 - 10.04 \quad \dots (i)$$



$$\text{pOH}_2 = \text{pKb} + \log \frac{[2]}{(40m - 2)} = 14 - 9.14 \quad \dots (ii)$$

eq. (ii)eq(i)

$$10.04 - 9.14 = \log \frac{[2]}{[40m-2]} \times \frac{[40m-0.5]}{[0.5]}$$

$$0.9 = \log 2^3 = \log \frac{4[40m-0.5]}{[40m-2]}$$

$$(40m-2)8 = 4(40m-0.8)$$

$$m = 0.1125$$

8. **Ans.(5.00)**

9. **Ans.(5.00)**

Compound which are aldehyde or give Aldehyde in basic medium (hemiacetal) give +ve Tollen test

10. **Ans.(4.00)**

a, c, d, e gives hydrogen gas

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Sol. $(\sim(p \vee q)) \vee (\sim p \wedge q) \equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q)$
 $\sim p \wedge (\sim q \vee q) \equiv \sim p \wedge t \equiv \sim p$

2. **Ans. (C)**

Sol. Let variance of first n positive integer be

$$\text{Var}(n) = \sigma^2$$

i.e. Variance of first n positive multiple of 5 will be

$$\text{Var}(5n) = 25\sigma^2$$

Variance of first n positive multiple of 10 will be $\text{Var}(10n) = 100\sigma^2$

$$\Rightarrow \frac{\text{Var}(5n)}{\text{Var}(10n)} = \frac{25\sigma^2}{100\sigma^2} = \frac{1}{4}$$

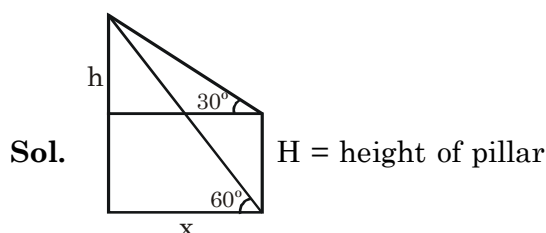
3. **Ans. (A)**

Sol. $|z+16|^2 = 16|z+1|^2$
 $\Rightarrow (z+16)(\bar{z}+16) = 16(z+1)(\bar{z}+1)$
 $\Rightarrow z\bar{z} + 16z + 16\bar{z} + 256$
 $= 16z\bar{z} + 16z + 16\bar{z} + 16$
 $\Rightarrow z\bar{z} = 16 \Rightarrow |z|^2 = 16 \Rightarrow |z| = 4$

4. **Ans. (C)**

Sol. $|a| \leq b$ and $|b| \leq c$
 $\Rightarrow |a| \leq c$
 $\Rightarrow$ relation is transitive

5. **Ans. (B)**



$$\tan 60^\circ = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}}$$

$$\tan 30^\circ = \frac{h-H}{x} \Rightarrow H = \frac{2h}{3}$$

6. **Ans. (B)**

Sol. We have A($\alpha, 6$), B(-5,0) and C(5,0)

If AB = AC, we get $\alpha = 0$.

Hence, we get one value of α

If AC = BC, we get $\alpha = -3, 13$

If AB = BC, we get $\alpha = 3, -13$

α can take 5 values : 0, 3, -3, 13, -13

7. **Ans. (A)**

Sol. Let the circle be $x^2 + (y-\alpha)^2 = a^2$. Let the point of intersection of tangents at P and Q be (h,k).

Then equation of PQ, is

$$hx + (k-\alpha)(y-\alpha) - a^2 = 0.$$

As PQ passes through (a,0), we have

$$ha - \alpha(k-\alpha) - a^2 = 0$$

$$\Rightarrow \alpha^2 - k\alpha + a(h-\alpha) = 0$$

Now for real values of α , we have $D \geq 0$.

$$\Rightarrow k^2 - 4a(h-a) \geq 0$$

$$\text{or } y^2 \geq 4a(x-a)$$

8. **Ans. (B)**

Sol. Let $A\left(ct_1, \frac{c}{t_1}\right), B\left(ct_2, \frac{c}{t_2}\right)$

equation of AB is

$$y = -\left(\frac{1}{t_1 t_2}\right)x + c\left(\frac{t_1 + t_2}{t_1 t_2}\right) \dots(1)$$

This three is tangent to $y^2 = 16x$

$$\Rightarrow c \left(\frac{t_1 + t_2}{t_1 t_2} \right) = \frac{4}{\left(-\frac{1}{t_1 t_2} \right)}$$

$$c(t_1 + t_2) = -4(t_1 t_2)^2 \quad \dots(2)$$

midpoint (x, y) of AB is

$$x = \frac{c(t_1 + t_2)}{2}, y = \frac{c(t_1 + t_2)}{2t_1 t_2}$$

eliminating t_1 and t_2

$$2x = -4 \left(\frac{y}{x} \right)^2$$

$\Rightarrow y^2 = -2x$ which is parabola whose latus rectum is 2.

9. **Ans. (A)**

Sol. Let M be $(\sqrt{18}\cos\theta, \sqrt{8}\sin\theta)$. As shortest distance lies along common normal Given line will be parallel to the tangent at M

$$\frac{x\cos\theta}{\sqrt{18}} + \frac{y\sin\theta}{\sqrt{8}} = 1 \text{ is parallel to}$$

$$2x - 3y + 25 = 0$$

$$\Rightarrow -\frac{\sqrt{8}\cos\theta}{\sqrt{18}\sin\theta} = \frac{2}{3} \Rightarrow \tan\theta = -1$$

$$\Rightarrow \theta = \frac{3\pi}{4} \Rightarrow \text{point M is } (-3, 2)$$

10. **Ans. (A)**

$$\text{Sol. } [\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}]$$

$$= |\vec{a} \vec{b} \vec{c}|^2 = |\vec{c} \cdot (\vec{a} \times \vec{b})|^2 = |\vec{c}|^2 |\vec{a} \times \vec{b}|^2 \cos^2 \theta$$

The maximum value of $|\vec{a} \vec{b} \vec{c}|$ is possible only when $\vec{c}$ is parallel to $\vec{a} \times \vec{b}$.

$$\theta = 0 \text{ or } \pi$$

$$\text{Hence, } |(\vec{a} \times \vec{b}) \times \vec{c}|^2 = |\vec{a} \times \vec{b}|^2 |\vec{c}|^2 \sin^2 \theta = 0.$$

11. **Ans. (D)**

Sol. Required plane is perpendicular to the line of intersection of given two planes. Vector along line of intersection of planes is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 3 & -4 & 1 \end{vmatrix} = 6\hat{i} + 2\hat{j} - 10\hat{k}$$

$$\therefore \text{Required plane is } 3x + y - 5z = 0$$

12. **Ans. (C)**

Sol. Sum = Product

$$\frac{4}{a} = a \Rightarrow a^2 = 4$$

$$\text{Also } D < 0 \Rightarrow (4)^2 - 4a^3 < 0$$

$$4(4 - a^3) < 0$$

$$a = 2$$

13. **Ans. (A)**

Sol. A.M of $1, 2, 3, \dots, n = \frac{n+1}{2}$

If $k, k+1$ are removed then

$$\text{AM} = \frac{\frac{n(n+1)}{2} - (2k+1)}{n-2}$$

$$\Rightarrow 1 = \frac{n+1}{2} - \frac{n(n+1) - 2(2k+1)}{2(n-2)} \Rightarrow n - k = 1$$

14. **Ans. (B)**

Sol. $x + y + z = -12$

$$\text{Here } -10 \leq x, y, z \leq -1$$

$$\text{Let } x_1 = -x, y_1 = -y, z_1 = -z$$

$$\therefore \text{ we have } x_1 + y_1 + z_1 = 12,$$

$$\text{where } 1 \leq x_1, y_1, z_1 \leq 10$$

$$\therefore \text{ Number of solutions of the equation } = {}^{12-1}C_{3-1} = {}^{11}C_2 = 55$$

15. **Ans. (B)**

Sol. General term of $\left(x - \frac{3}{x}\right)^{11}$

$$T_{r+1} = {}^{11}C_r x^{11-r} \left(-\frac{3}{x}\right)^r = {}^{11}C_r x^{11-2r} (-3)^r$$

$11 - 2r \neq 0, -2$ so $11 - 2r = -1$ is only possible case

$$\therefore r = 6$$

$$\therefore \text{ coefficient is } (-3)^6 {}^{11}C_6$$

16. **Ans. (C)**

Sol. Following cases are possible

Red	Blue	Green
2	1	1
3	1, 2	1, 2
4	1, 2, 3	1, 2, 3
5	1, 2, 3, 4	1, 2, 3, 4
6	1, 2, 3, 4, 5	1, 2, 3, 4, 5

Required probability

$$= \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{2}{6} \times \frac{2}{6} + \frac{1}{6} \times \frac{3}{6} \times \frac{3}{6} + \frac{1}{6} \times \frac{4}{6} \times \frac{4}{6} + \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} = \frac{55}{216}$$

17. Ans. (C)

$$\begin{aligned}\text{Sol. } f(\theta) &= 1 + \sin^2\theta - \left(\cos^2\left(\theta + \frac{2\pi}{3}\right) - \sin^2\left(\theta + \frac{4\pi}{3}\right) \right) \\ &= 1 + \sin^2\theta - \left\{ \cos(2\theta + 2\pi) \cos\left(\frac{2\pi}{3}\right) \right\} \\ &= 1 + \sin^2\theta + \frac{1}{2}(1 - 2\sin^2\theta) = \frac{3}{2} \\ \Rightarrow f\left(\frac{\pi}{8}\right) + f\left(\frac{\pi}{12}\right) &= 3\end{aligned}$$

18. Ans. (A)

$$\begin{aligned}\text{Sol. } \sec x + \operatorname{cosec} x &= \sin x + \cos x \\ \Rightarrow \frac{\sin x + \cos x}{\sin x \cos x} &= \sin x + \cos x \\ \Rightarrow (\sin x + \cos x)(1 - \sin x \cos x) &= 0 \\ \Rightarrow \sin x = -\cos x \text{ or } 1 &= \sin x \cos x \\ \Rightarrow \tan x = -1 \text{ or } \sin 2x &= 2 \text{ not possible} \\ \Rightarrow x = n\pi - \frac{\pi}{4}, x \in I &\Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}\end{aligned}$$

19. Ans. (C)

$$\begin{aligned}\text{Sol. } \int (2 - x \tan x) \sqrt{\cos x} dx \\ &= \int (2\sqrt{\cos x} - x \tan x \sqrt{\cos x}) dx \\ &= \int (f(x) + x f'(x)) dx \\ &= x f(x) = 2x\sqrt{\cos x} + C\end{aligned}$$

20. Ans. (B)

Sol. By dividing both the sides by $\cos^2 y$. We get

$$\sec^2 y \frac{dy}{dx} + \frac{1}{x} 2 \tan y = x^3$$

$$\text{Put } \tan y = z \Rightarrow \sec^2 y \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \frac{dz}{dx} + \frac{2}{x} z = x^3, \text{ which is linear in } z$$

The integrating factor is

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} = x^2$$

Hence, the solution is

$$z(x^2) = \int x^3(x^2) dx \Rightarrow zx^2 = \frac{x^6}{6} + a$$

$$\therefore (6 \tan y) x^2 = x^6 + 6a$$

$$\Rightarrow 6x^2 \tan y = x^6 + c, [c = 6a]$$

1. Ans. 13.00

$$\text{Sol. } g'(y) = \frac{1}{f'(x)} = \frac{1}{6x^2 + 7}$$

$$x = 1$$

$$y = 4$$

$$g'(4) = \frac{1}{13}$$

2. Ans. 6.00

$$\text{Sol. } L = \lim_{\theta \rightarrow 0} \frac{2(\ln \sin \theta - \ln \theta)}{\theta^2}$$

$$(\text{by putting } x = \frac{1}{\theta^2})$$

$$= \lim_{\theta \rightarrow 0} \frac{\cot \theta - \frac{1}{\theta}}{\theta} \quad (\text{L'H Rule})$$

$$= \lim_{\theta \rightarrow 0} \frac{\theta - \tan \theta}{\theta^2 \tan \theta} = \lim_{\theta \rightarrow 0} \frac{\theta - \left(\theta + \frac{\theta^3}{3} + \dots \right)}{\theta^3 \times \left(\frac{\tan \theta}{\theta} \right)} = -\frac{1}{3}.$$

3. Ans. 16.00

$$\text{Sol. } f(x) = \sin^{-1} \left(\frac{1-x}{1+x} \right)$$

$$\frac{df}{dx} = \frac{1}{\sqrt{1 - \left(\frac{1-x}{1+x} \right)^2}} \left(\frac{-(1+x) - (1-x)}{(1+x)^2} \right)$$

$$\frac{df}{dx} = \left(\frac{|1+x|}{\sqrt{4x}} \right) \frac{-2}{(1+x)^2}$$

$$g(x) = \sqrt{x}$$

$$\frac{dg}{dx} = \frac{1}{2\sqrt{x}}$$

Derivative of $f(x)$ wrt $g(x)$

$$\frac{df}{dg} = \frac{\frac{df}{dx}}{\frac{dg}{dx}} = \frac{-2|1+x|}{(1+x)^2}$$

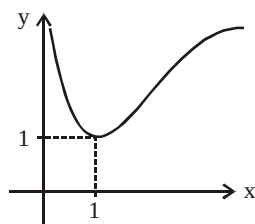
$$p = -\frac{4}{3} \text{ if } x = \frac{1}{2}$$

4. Ans. 1.00

Sol. $x > 0$ (Domain)

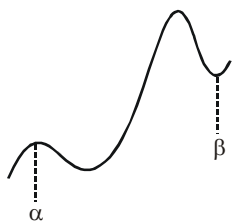
$$\therefore k = \ln x + \frac{1}{x} = f(x)$$

Graph of $f(x)$:



5. Ans. 4.00

Sol. $f''(\alpha) = -ve$ and $f''(\beta) = +ve$
so $x = \alpha$ is a point of maxima
& $x = \beta$ is point of minima



$\Rightarrow$ minimum 4 roots of $f'(x) = 0$ in $[\alpha - 1, \beta + 1]$

6. Ans. 1.00

$$\text{Sol. } I = \int_{\pi/2}^0 \frac{e^{\tan^{-1}(\sin x)}}{e^{\tan^{-1}(\sin x)} + e^{\tan^{-1}(\cos x)}} dx + \int_0^{5\pi/2} \frac{e^{\tan^{-1}(\sin x)}}{e^{\tan^{-1}(\sin x)} + e^{\tan^{-1}(\cos x)}} dx = I_1 + I_2$$

$$I_1 = -\frac{\pi}{4} \text{ and } I_2 = \frac{5\pi}{4} \text{ (Using king)}$$

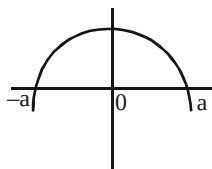
$$I = \pi \Rightarrow k = 1$$

7. Ans. 4.00

Sol. $ay = 3(a - x)(a + x)$

$$\text{Area} = 2 \int_0^a \frac{3}{a} (a^2 - x^2) dx = 64$$

$$\Rightarrow \frac{3}{a} \left(\left(a^2 x - \frac{x^3}{3} \right) \right)_0^a = 32$$



$$\Rightarrow \frac{3}{a} \left(a^3 - \frac{a^3}{3} \right) = 32 \Rightarrow 3 \left(a^2 - \frac{a^2}{3} \right) = 32$$

$$\Rightarrow 3a^2 - a^2 = 32$$

$$\Rightarrow 2a^2 = 32 \Rightarrow a^2 = 16 \Rightarrow a = \pm 4$$

8. Ans. 3.00

$$\begin{aligned} \text{Sol. } \sum_{r=1}^n \tan^{-1} \left(\frac{2}{(r+1)^2} \right) &= \tan^{-1} \left(\frac{(r+2)-r}{1+r(r+2)} \right) \\ &= (\tan^{-1} 3 - \tan^{-1} 1) + (\tan^{-1} 4 - \tan^{-1} 2) \\ &\quad + (\tan^{-1} 5 - \tan^{-1} 3) + \dots + (\tan^{-1}(n+1) - \tan^{-1}(n-1)) + (\tan^{-1}(n+2) - \tan^{-1} n) \\ &= \tan^{-1}(n+1) + \tan^{-1}(n+2) - \tan^{-1} 1 - \tan^{-1} 2 \\ \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{2}{(r+1)^2} \right) &= \frac{\pi}{2} + \frac{\pi}{2} - \tan^{-1} 1 - \tan^{-1} 2 \\ &= \tan^{-1} 3 \end{aligned}$$

9. Ans. 1.00

$$\text{Sol. } ||A^{-1}|(AB)^T| = ||A^{-1}| \cdot (A \cdot \text{adj} A)^T|$$

$$= \left| \frac{1}{|A|} \cdot (|A| \cdot I)^T \right| = \left| \frac{1}{|A|} \cdot |A| \cdot I^T \right|$$

10. Ans. 5.00

Sol. Given the system of equation

$$x + y - z = 2 \quad \dots(1)$$

$$2x - y + 4z = 1 \quad \dots(2)$$

$$x + \alpha y + z = \beta \quad \dots(3)$$

solving (1) and (2) : $x = 1 - z, y = 1 + 2z$

$$\therefore (3) \Rightarrow 1 - z + \alpha(1 + 2z) + z = \beta$$

$$\Rightarrow 2\alpha z = \beta - \alpha - 1$$

$\therefore$ For unique solution $\alpha \neq 0, \beta \in \mathbb{R}$

$$\therefore A = 4 \times 5 = 20$$

For no solution $\alpha = 0, \beta \neq 1$

$$\Rightarrow B = 1 \times 4 = 4$$

and for infinite number of solutions,

$$\alpha = 0, \beta = 1 \Rightarrow C = 1$$

$$\therefore A - 4B + C = 20 - 16 + 1 = 5.$$